TENOFOVIR

Do not stop suddenly in HBV

Monitor kidneys & bone

CHEMICAL NAME Tenofovir disoproxil fumarate: bis{[(isopropoxycarbonyl)oxy]methyl} ({[(2R)-1-(6-amino-9H-purin-9-yl)propan-2-yl]oxy}methyl)phosphonate fumarate (TDF); also tenofovir alafenamide (TAF)

CLASS Nucleotide reverse transcriptase inhibitor (NtRTI)

TRADE NAMES Viread (TDF) · Vemlidy (TAF) · Truvada (with emtricitabine) · Tenvir

THERAPEUTIC USES / INDICATIONS

- Chronic hepatitis B (first-line).
- HIV-1 infection (with other antiretrovirals).
- HIV pre-exposure prophylaxis (PrEP) (with emtricitabine).
- Prevention of mother-to-child HBV transmission (high viral load, 3rd trimester).

ADMINISTRATION

- **PO** once daily at the same time; TDF with or without food (better with food); TAF with food.
- Never as monotherapy for HIV.

DOSAGES

- **TDF:** 300 mg once daily.
- **TAF (HBV):** 25 mg once daily.
- **PrEP:** 1 tablet (TDF 300 mg/FTC 200 mg) daily.
- TDF: CrCl 30–49 every 48 h; avoid <30 (use TAF or adjust per guideline).

PHARMACOKINETICS

- **Peak:** 1–2 h. Prodrug converted intracellularly to tenofovir diphosphate.
- Not metabolised by CYP.
- Excreted unchanged in urine. Intracellular **t½:** >60 h.

PHARMACODYNAMICS

- Tenofovir diphosphate competes with natural nucleotides and terminates viral DNA chain by inhibiting **reverse transcriptase** (HIV) and **HBV polymerase**.
- High barrier to HBV resistance.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea, flatulence.
- Headache, fatigue.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (Fanconi syndrome, proximal tubulopathy) — mainly TDF.
- Loss of bone mineral density, osteomalacia (TDF).
- Lactic acidosis with hepatomegaly (rare).
- Severe hepatitis flare on stopping in HBV.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** renal impairment, osteoporosis, use with other nephrotoxic drugs.

KEY INTERACTIONS

- **NSAIDs (high dose, multiple), aminoglycosides:** ↑ nephrotoxicity.
- **Didanosine:** ↑ toxicity.
- **Ledipasvir/sofosbuvir:** ↑ tenofovir levels.

NURSING MANAGEMENT

- Baseline and regular creatinine, phosphate, urinalysis (protein, glucose).
- Test for HIV before HBV treatment and HBV before HIV treatment.
- Monitor LFTs, HBV DNA, HIV viral load.
- Reinforce strict adherence.
- PrEP: HIV test every 3 months.

PATIENT EDUCATION

- Take every day — missing doses leads to resistance.
- Do not stop without your doctor (HBV flare can be serious).
- Report bone pain, muscle weakness, increased urination.
- PrEP does not prevent other STIs; use condoms.

KEY POINTS

- NtRTI for HBV, HIV and PrEP.
- Stopping can flare hepatitis B.
- TDF: kidney and bone toxicity (less with TAF).
- Check renal function regularly.